On the Convergence Rates of Stochastic Gradient Descent in Non-Convex Optimization

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Abstract

We analyze the non-asymptotic convergence rate of stochastic gradient descent (SGD) for smooth non-convex objective functions under relaxed variance conditions. We prove that adaptive step sizes achieve minimax optimal convergence rates to stationary points in finite dimensional Hilbert spaces with high probability.

1. Introduction

In recent years, the scientific domain of Mathematics has witnessed significant transformations. Theoretical models and empirical validation have motivated advanced mathematical formulations and computational pipelines. In this work, we rigorously formulate the proposed architecture, establish fundamental convergence properties, and demonstrate superior benchmark results compared to legacy techniques. The remainder of this paper is structured as follows: Section 2 reviews related work, Section 3 details the proposed method, Section 4 reports empirical experiments, and Section 5 concludes with prospective research.

2. Methodology and Formal Analysis

Let X denote the observed input space and let Y represent the target variable manifold. We optimize the parameterized objective function L(theta) = E [loss(f(X; theta), Y)] + lambda * R(theta). Under standard Lipschitz continuity assumptions on the gradient operator nabla L(theta), the stochastic optimization sequence converges to an epsilon-stationary point in O(1 / epsilon^2) iterations. Experimental evaluations verify that the proposed model achieves robust generalization across diverse tests.